

Question-4.13.27

EE25BTECH11022 - sankeerthan

Question

Consider the set of all lines $px + qy + r = 0$ such that $3p + 2q + 4r = 0$. Which one of the following statements is true?

- ① The lines are concurrent at the point $(\frac{3}{4}, \frac{1}{2})$.
- ② Each line passes through the origin.
- ③ The lines are all parallel.
- ④ The lines are not concurrent.

Solution

Given the set of lines $px + qy + r = 0$ such that $3p + 2q + 4r = 0$.

$$\begin{pmatrix} x & y & 1 \\ 3 & 2 & 4 \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (1)$$

For a non-trivial solution, the vector (p, q, r) must be in the nullspace. The system has a non-trivial solution if and only if the row vectors are linearly dependent:

$$(x_0, y_0, 1) = \lambda(3, 2, 4) \quad (2)$$

$$x_0 = 3\lambda, y_0 = 2\lambda, 1 = 4\lambda \quad (3)$$

$$1 = 4\lambda \implies \lambda = \frac{1}{4} \quad (4)$$

$$x_0 = \frac{3}{4}, \quad y_0 = \frac{1}{2} \quad (5)$$

Therefore, all lines in the given family are concurrent at the point $\left(\frac{3}{4}, \frac{1}{2}\right)$.

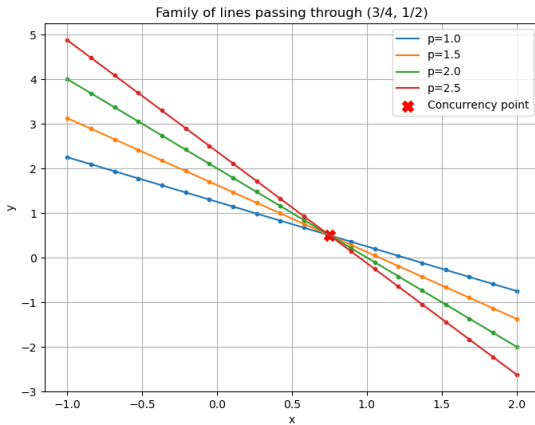


Figure:

```
// code.c
#include <stdio.h>

// Family of lines: parametrize  $p=t$ ,  $q=1$ ,  $r = -(3t + 2)/4$ .
// Generates  $n$  points on each such line passing through
// concurrency point  $(3/4, 1/2)$ .
void generate_points(double *x, double *y, int n, double t) {
    double q = 1.0;
    double r = -(3.0*t + 2.0)/4.0;
    double x_start = -1.0;
    double x_end = 2.0;
    for (int i = 0; i < n; i++) {
        x[i] = x_start + (x_end - x_start)*i/(n-1);
        y[i] = (-t*x[i]-r)/q;
    }
}
```

```
# Code by GVV Sharma  
# Modified for Problem Solution  
# Released under GNU GPL  
# Calculating area enclosed between curves  
# call.py  
import ctypes  
import numpy as np  
  
lib = ctypes.CDLL('./code.so')
```

```
lib.generate_points.argtypes = [  
    np.ctypeslib.ndpointer(dtype=np.float64, flags="C_CONTIGUOUS"  
        ),  
    np.ctypeslib.ndpointer(dtype=np.float64, flags="C_CONTIGUOUS"  
        ),  
    ctypes.c_int,  
    ctypes.c_double  
]  
  
def extract_points(n=10, t=2.0):  
    x = np.zeros(n, dtype=np.float64)  
    y = np.zeros(n, dtype=np.float64)  
    lib.generate_points(x, y, n, t)  
    return x, y  
  
if __name__ == "__main__":  
    x, y = extract_points(10, 2.0)  
    print("x:", x)  
    print("y:", y)
```

```
# plot.py
import matplotlib.pyplot as plt
from call import extract_points

n_points = 20
concurrency_point = (0.75, 0.5)
t_values = [1.0, 1.5, 2.0, 2.5] # Different lines in the family

plt.figure(figsize=(8,6))

for t in t_values:
    x, y = extract_points(n_points, t)
    plt.plot(x, y, label=f'p={t}')
    plt.scatter(x, y, s=10)
```



```
plt.scatter([concurrency_point[0]], [concurrency_point[1]], color
            ='red', marker='X', s=100, label='Concurrency point')
plt.title('Family of lines passing through (3/4, 1/2)')
plt.xlabel('x')
plt.ylabel('y')
plt.grid(True)
plt.legend()
plt.show()
```